

Recap Checkpoint 6 — Full Year 12 Cumulative (Pre-Mock): 1.1, 1.3–1.5 + 2.1–2.3

Name: _____ Date: _ **Mark:** _____ / 46

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). (1.1 — Processors) State **three** factors that affect the **performance** of a processor, and for each give a brief explanation of its effect.

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Q2 [4] (AO2). (1.4 — Number bases) Show all working.

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(a) Convert denary **181** to **8-bit binary**. [1]

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(b) Convert **181** to **hexadecimal**. [2]

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(c) Convert binary **1010 0110** to denary. [1]

Q3 [4] (AO2). (1.4 — Two's complement) Using **8-bit two's complement**, calculate $-40 + 25$.

(a) Show 40 and 25 in 8-bit binary. [1]

(b) Form the two's complement representation of -40 . [1]

(c) Perform the addition and give the final **denary** result. [2]

Q4 [3] (AO2). (1.4 — Boolean algebra / floating point)

(a) State why floating-point numbers are **normalised**. [1]

(b) Simplify $A + A \cdot B$ and name the law used. [2]

Q5 [4] (AO1). (1.3 — Networks)

(a) Explain the difference between **circuit switching** and **packet switching**. [2]

(b) State **two** items of data added to a packet's **header**. [2]

Q6 [3] (AO1/AO3). (1.5 — Legal/ethical) Self-driving cars must sometimes "decide" how to act in an unavoidable collision.

Discuss **one ethical** issue this raises and identify **who** might be held **legally responsible** in a crash. [3]

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Q7 [4] (AO2). (2.1 / 2.2 — *Computational thinking + programming*)

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(a) State the difference between a **procedure** and a **function**. [1]

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(b) Explain the difference between **passing a parameter by value** and **by reference**. [2]

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(c) State **one** advantage of using **local** variables over **global** variables. [1]

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Q8 [5] (AO2). (2.3 — *Algorithm trace*) A list contains [7, 2, 9, 4, 1]. Carry out a **bubble sort**, showing the list after **each complete pass**. State how many passes are needed before the list is sorted, and give the **worst-case** time complexity of bubble sort.

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Q9 [4] (AO1/AO2). (2.3 — *Big O reasoning*)

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(a) State the worst-case Big O of: insertion sort, merge sort, and a binary search. **[3]**

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(b) Explain why merge sort is preferable to insertion sort for very large datasets, referencing growth rates. **[1]**

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Q10 [6] (AO3). (2.3 — *Graphs, applied*) Using **Dijkstra's algorithm**, find the shortest **distance** and shortest **path** from **A** to **E** in the weighted, undirected graph below. Show a table of tentative distances and the order in which nodes are visited.

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Edges (undirected, weighted):

$A-B = 2$, $A-C = 5$, $B-C = 1$, $B-D = 7$, $C-D = 3$, $C-E = 8$, $D-E = 2$

Q11 [6] (AO3). (*Synoptic extended response*) A hospital is replacing a paper appointment system with networked software that stores patient data, lets staff search records, and sends automated reminders.

In an extended response, discuss the design considering: **(i)** how **abstraction and decomposition** help manage the build, **(ii)** an appropriate **data structure or search algorithm** for fast record lookup (justify with time complexity), and **(iii)** **one legal** consideration for storing patient data. **[6]**
